Revisiting the Fundamentals of Swarm Robotics: Event-based Analysis of Biological Swarms

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***Abstract*— Scalability and high-speed coordination remain primary challenges in aerial swarm robotics even though the agility of individual autonomous aerial robots has increased dramatically in the recent years. To tackle these challenges, we revisit some of the core methods and assumptions commonly applied in swarm robotics by studying the underlying principles of the biological systems that inspire and motivate most of the research in this field. Specifically, we analyze maneuvers of large-scale flocks of *Columba livia* (rock dove) using data captured by an event stereo camera. This setup provides novel insights over existing research relying on standard cameras or Global Navigation Satellite Systems (GNSS) trackers, such as 3D trajectory reconstruction with a microsecond-accurate temporal resolution or even wingbeat frequency and other visual cues that may be utilized by the animals for agile coordination. Our analysis shows that these biological flocks utilize a dynamic leadership mechanism driven by local visual perception, which challenges the classical consensus-based robotic approaches such as boids-like swarming. As we show in simulated experiments, a dynamic leadership-based swarming rule not only better explains the observed biological data when compared to a conventional approach, but also leads to more efficient trajectories of the individual agents. We publish details about our methodology including the design of the event stereo camera device for capturing the data, open-source implementation, as well as the datasets, to stimulate further research of biological swarms and the transfer of these findings to robotic systems.**

***Index Terms*— Bio-inspired Robot Learning, Swarm Robotics, Multi-Robot Systems**

Supplementary Material

The dataset, code, labels, and additional resources are available online at The dataset, code, labels, and additional resources are available online at https://bit.ly/4rPcpHV.

I. Introduction

Although agility of individual autonomous Unmanned Aerial Vehicles (UAVs) has advanced significantly in the recent years, coordinated groups of aerial robots are still lagging behind their biological counterparts, such as bird flocks capable of fast collective maneuvers in cluttered environments without explicit communication [1]. Robotics researchers have traditionally taken inspiration from such biological systems, but the actual understanding of their underlying principles is limited. In this paper, we show that

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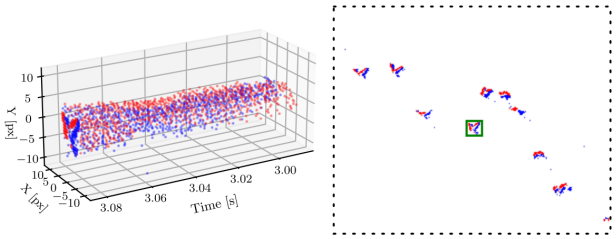

Fig. 1: Microsecond-resolution event tracking of a flock. *Left:* Point-cloud representation of events generated by a single bird over 80 ms. *Right:* Visualization of the corresponding event image.

the mentioned limitations of aerial robotic swarms may be mitigated by understanding these animal behaviors in more depth.

To this end, we present a new methodology for studying the collective motion of birds using a stereo event camera (see Fig. 1). Event-based vision enables extracting information that was previously unavailable and at a high temporal resolution, such as 3D trajectories of individuals within a large-scale flock or detailed visual cues that may be used by the animals for implicit agile coordination. Utilizing this methodology, we have captured several sequences of flocks of *Columba livia* (rock dove, also referred to as the common pigeon) during collective maneuvers at an unprecedented scale (the largest analyzed flock has over 80 individuals) and temporal resolution (10 µs). Our analysis of this data shows that the birds follow a dynamic leadership flocking rule, which challenges the classical and widely adopted consensus-based swarming algorithms [2]–[4], and supports the various approaches based on dynamic hierarchical reorganization, such as [5]–[7]. To evaluate the application of these insights to multi-robot systems, we compare a conventional boids-like model, a fixed leadership approach, and a dynamic leadership model inspired by the observed behavior. The results demonstrate that the dynamic leadership model better fits the trajectories of the observed birds, and also leads to a reduction in trajectory length and duration, increasing the swarm's efficiency.

II. Related Work

The classical consensus-based flocking models largely rely on the seminal "boids" framework [4], which was designed to visually imitate biological swarms, and which formulates the problem based on inter-agent distances and assumes omniscient knowledge of agents' states. We look to biological systems not for aesthetic fidelity but for functional efficiency. Unlike robots, birds do not possess internal distance sensors or communication networks. They navigate using local

visual cues [8] and perceive aggregate density rather than tracking discrete individuals [9]. By shifting from metric to perception-based salience, the loop can be significantly tightened.

A. Hierarchical Structure of Bird Flocks

Previous studies have extensively analyzed collective behavior, proposing various hypotheses for hierarchical leadership in small pigeon flocks [10], [11]. However, capturing how these leadership roles dynamically shift during rapid maneuvers in large flocks remains challenging with traditional sensors. We aim to empirically validate these transient leadership mechanisms using our microsecond-resolution, event-based tracking methodology to capture the fast visual cues underlying the kinematic interaction within the swarm.

B. Data Acquisition: GNSS vs. Vision

High-precision Global Navigation Satellite Systems (GNSS) logging devices have been instrumental in revealing flock structures [12]–[14]. However, their physical payload and the logistical intractability of tagging every individual make them unscalable for large flocks. Furthermore, GNSS data is purely metric. It records absolute positions but fails to capture the visual cues driving collective behavior, and low level dynamics like wing beats.

To bridge this sensory gap, computer vision approaches are essential. For instance, multi-camera systems have been used to reconstruct 3D trajectories of individuals within massive starling flocks [15]. While this demonstrated the viability of vision for capturing dense collective behavior, the analysis primarily centered on structural correlations rather than isolating local behavioral mechanisms.

Recently, [16] introduced "FlockSeer", a portable system using smartphone cameras to track birds. While it scales well for monitoring various individuals, its reliance on RGB sensors results in significant data redundancy, storage bottlenecks, and limits the observable dynamics. By contrast, the asynchronous nature of event-based sensing captures only the necessary motion data, providing a more efficient way to analyze the high-density interactions of large flocks, while providing a significantly higher temporal resolution.

C. Contributions

With respect to the related works, we claim the following contributions:

- Design of a stereo event-camera device and an automated pipeline integrating deep learning detection with multi-view data association for reconstructing 3D trajectories of individual agents within high-density flocks.
- A novel dataset of large-scale pigeon flocks with a high temporal resolution captured in the wild using the proposed system.
- An analysis of the captured data demonstrating that pigeons rely on a dynamic leadership mechanism to coordinate movement.
- A comparative analysis showing that dynamic leadership, rather than classical consensus-based averaging, is required to replicate the agility observed in natural flocks.

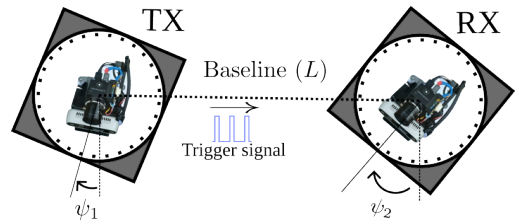

Fig. 2: The proposed device for capturing bird flock data comprising two time synchronized multi-sensor modules.

We release the recorded data, the implementation, and the hardware documentation as open source, to support future quantitative behavioral analysis, as we believe that gaining a deeper understanding of the rules and interactions governing biological flocks is crucial to advance the capabilities of multi-robot systems.

III. System Design

To capture dynamics of pigeon flocks, a custom stereoscopic system was developed. The hardware architecture, synchronization logic, and calibration procedures are detailed in this section.

A. Hardware Architecture

The experimental setup consists of two identical multi-sensor modules forming a stereo baseline (Fig. 2). Each module integrates three primary components:

1) **Event camera** (Prophesee EVK4 HD Event Camera): This serves as the primary sensor for behavioral analysis.
2) **RGB camera** (Intel RealSense D435): Included primarily for system redundancy, this sensor provides high-fidelity color context that serves as a visual baseline.
3) **Inertial Measurement Unit (IMU)** (BMI270): Rigidly attached to the event camera, this sensor provides gravity alignment and ensures a precise extrinsic relationship.

B. Synchronization Strategy

A simple software synchronization between the two independent sensor modules is insufficient for stereo reconstruction at the considered temporal resolution. To ensure time alignment, a hardware-level protocol is implemented based on a *Transmitter-Receiver* architecture. A microcontroller with an accurate clock source on the *Transmitter* acts as the primary time base. It generates a hardware trigger signal at 1 MHz that drives the local event camera and is simultaneously propagated via a differential driver over a Unshielded Twisted Pair (UTP) cable to the *Receiver* mother board. This ensures that the internal timestamp counters of both sensors are locked to the same reference with sub-microsecond precision.

TABLE I: Summary of the captured data sequences.

ID	Mode	Sensors active	Duration	Num. clips
1	Mono	Ev. + IMU + RGB	685 s	22
2	Mono	Ev. + IMU + RGB	1318 s	20
3	Stereo	Ev. + IMU (Left+Right)	754 s	22

C. Calibration Procedure

In order to accurately reconstruct full 3D trajectories of the birds, precise knowledge of both intrinsic and extrinsic parameters of the stereo camera system is required.

1) Intrinsic Calibration: The standard pinhole camera model with barrel distortion is employed for all sensors. While the calibration procedure typically utilizes several images of checkerboard or other pattern with known physical dimensions, event cameras require relative motion to perceive the pattern. To accommodate this, the checkerboard is moved to trigger event generation, and these events are integrated into accumulated frames. Standard corner detection is then applied to these reconstructed images to identify the checkerboard features, enabling the identification of focal lengths (f_x, f_y), principal points (c_x, c_y) and distortion coefficients.

2) Extrinsic Calibration: Determining the relative pose between the two stereo modules is critical. An approach combining physical measurement and inertial sensing is devised:

- **Orientation:** The pitch and roll angles of each camera were obtained directly from the gravity vectors measured by the IMUs onboard. The yaw is not observable via this method accelerometers, and was measured using a custom-built goniometric rig with 1° resolution placed on each module and a taut line stretched between the two modules (see Fig. 2). This approach is inspired by [16], which also provides a theoretical analysis of the accuracy of such calibration.
- **Translation:** The baseline distance was measured physically using a tape measure with centimeter resolution.

This calibration method ensures that the event-based stereo system is not only calibrated but also aligned with the global gravity frame, simplifying the subsequent data processing.

IV. Data Collection

The data acquisition was conducted at coordinates [redacted]. The experimental sessions took place during the Autumn of 2025.

A. Flight Protocol

To maximize the duration of data with the flock within the sensor's field of view, we exploited the characteristic strong site fidelity of pigeons. Two primary landing zones frequently used by the flock were identified. To stimulate the flock, acoustic and visual perturbations were applied at these two sites. This disturbed the birds and resulted in a take-off of the roof, followed by several minutes of circulating flights. The flock engaged in a "shuttling" behavior, flying repeatedly between the two landing zones before resting. The sensor array was positioned at the midpoint of this trajectory. This enabled the capture of extended sequences of uninterrupted flight, including straight-line transit, banking turns, and aborted landing maneuvers, which are rare in unprovoked settings.

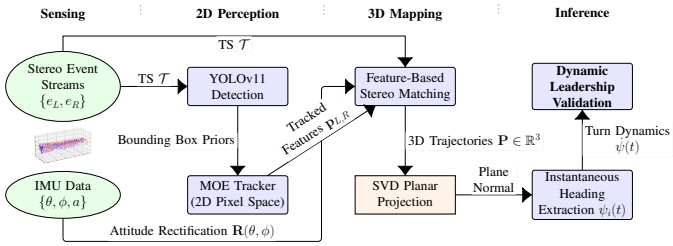

Fig. 3: System architecture for dynamic leadership validation using event-based vision.

B. Data Processing and Organization

The dataset is structured into three main sequences, organized into two validation phases based on the used sensor modalities to evaluate both monocular event processing and full stereoscopic reconstruction. To streamline data management and facilitate the analysis of specific behaviors, continuous event recordings were segmented into a series of discrete clips, as summarized in Table I. These segments range from 4 to 45 seconds, with their duration defined strictly by the interval in which the flock remains within the sensors' Field Of View (FOV). Any periods where birds were outside the FOV or were not flying were removed to ensure the resulting dataset contains only active flight maneuvers.

1) Phase I: Multimodal Baseline (Seq. 1 & 2): Recorded at 09:38 and 10:33, these sequences utilize a single sensor module with the full sensor suite: event camera, IMU, and RGB camera. The synchronized RGB frames were used for validation prior to stereoscopic deployment.

2) Phase II: Event-Based Stereo (Seq. 3): Recorded at 11:59, this phase represents the core experimental contribution. In this sequence, the redundant RGB acquisition was disabled due to storage and technical limitations. Initial sequences revealed a data volume difference of nearly an order of magnitude: *Sequence 1* required 15.2 GiB for events versus 116.4 GiB for RGB data, while *Sequence 2* generated 29.1 GiB versus 233.5 GiB. Beyond the extreme storage requirements, the high bandwidth of the RGB sensors caused connectivity issues during field deployment. Relying exclusively on event data provided a more stable and scalable system even for resource-constrained platforms, which is important in field deployments.

V. 3D Stereo Event Matching

This section describes the pipeline for extraction of 3D trajectories of individual birds from the stereo event streams captured by the stereo device. These trajectories are then analyzed and used to validate the dynamic leadership hypothesis (as described in section VI). The whole data processing pipeline is presented in Fig. 3.

A. Event Representation and Object Detection

Event cameras produce a stream of discrete events $e_k = (x_k, y_k, t_k, p_k)$, where x_k, y_k are pixel coordinates, t_k is the timestamp and p_k the polarity of the event. To leverage

(a) TS representation. (b) EVF representation.

Fig. 4: Bird detections on different event representations. The green dots indicate the calculated center of mass for both Time Surface (TS) and EVent Frames (EVF) representations.

the spatial feature extraction capabilities of conventional computer vision methods such as Convolutional Neural Networks (CNNs), this sparse spatiotemporal stream is projected into a dense tensor format (i.e. an image). Two distinct representation methods were evaluated w.r.t. the detection accuracy.

First, the EVent Frames (EVF) representation [17] is generated by converting events within a defined temporal window into a 2D image through pixel-wise polarity accumulation. Second, the Time Surface (TS) representation [18] captures the recent dynamic history of the scene by applying an exponential decay kernel to incoming events to create a richer spatio-temporal context. Both representations are visualized in Fig. 4.

1) Detector Performance Comparison: The YOLOv11 [19] architecture was trained on a manually labeled subset using a short temporal window of $\Delta t = 5\,\text{ms}$ for each label to isolate fast motion and preserve distinct event morphology.

As presented in Table II, the TS representation yields superior overall performance, achieving a 1.2 % improvement in mean Average Precision (mAP)@0.5 compared to EVF. A stability analysis of the 10 final training epochs indicates that EVF reaches a marginally more stable steady state. However, the TS representation was ultimately selected for the full pipeline because maximizing detection accuracy was prioritized over minimal variance. In the context of this application, the improvements in correctly identifying and localizing the targets outweigh the minor fluctuations observed during the final training stages.

B. Multi-Object Event (MOE) Tracking Methodology

Standard tracking algorithms struggle with the specific challenges presented by the considered datasets, such as extreme visual similarity between individual tracked objects, high object density, small sizes of the objects, and occlusions and overlaps of the objects' trajectories. To address these limitations, a custom MOE tracking methodology is proposed. The performance of this system is evaluated against established baseline trackers to validate its effectiveness in maintaining robust tracking in these challenging scenarios.

Let $\mathcal{D}_t = \{\mathbf{d}_t^1, \mathbf{d}_t^2, \ldots, \mathbf{d}_t^M\}$ be the set of detections provided by the YOLOv11 model from the event frame representation at time-step t, where each detection $\mathbf{d}_t^j \in \mathbb{R}^2$ represents the centroid coordinates (x, y) of the j-th bird. Let us denote the set of active tracks as $\mathcal{T}_{t-1} = \{T_{t-1}^1, \ldots, T_{t-1}^N\}$. The i-th track maintains a state vector $S_t^i = (\mathbf{p}_t^i, \mathbf{v}_t^i)$, where $\mathbf{p}_t^i$ denotes the position and $\mathbf{v}_t^i$ denotes the velocity vector.

TABLE II: YOLOv11 detection performance with different event representations.

Method	Prec.	Recall	mAP@0.5	mAP@0.5 (σ)
EVF	0.9064	0.88755	0.93521	**0.0023**
TS	**0.9270**	**0.90745**	**0.94737**	0.0044

1) Motion Prediction: Assuming a linear constant-velocity model, before receiving new frames at time-step t, the predicted position $\hat{\mathbf{p}}_t^i$ for each existing track is calculated as $\hat{\mathbf{p}}_t^i = \mathbf{p}_{t-1}^i + \Delta t \mathbf{v}_{t-1}^i$, where $\Delta t = 5\,\text{ms}$.

2) Data Association and Outlier Rejection: To associate existing tracks with new detections, we construct a cost matrix based on the Euclidean distance between the predicted state $\hat{\mathbf{p}}_t^i$ and the set of measured centroids $\mathbf{d}_t^j$. For a track i and a detection j, the spatial cost C_{ij} is defined as $C_{ij} = \|\hat{\mathbf{p}}_t^i - \mathbf{d}_t^j\|_2$.

The association problem is solved using a greedy matching strategy subject to a spatial gating threshold θ_{gate}. A match (i, j) is considered valid if $C_{ij} < \theta_{\text{gate}}$, where $\theta_{\text{gate}} = 8.65\,\text{px}$.

To further improve robustness against false positives and erratic detections, a dynamic velocity gating mechanism is introduced based on the collective motion of the flock. The mean flock magnitude velocity $\bar{v}_t$ is calculated, as

$$\bar{v}_t = \frac{1}{|\mathcal{T}_{t-1}|} \sum_{k=1}^{|\mathcal{T}_{t-1}|} \|\mathbf{v}_{t-1}^k\|_2, \tag{1}$$

A detection $\mathbf{d}_t^j$ associated with track i is rejected as an outlier if the condition

$$\|\mathbf{d}_t^j - \mathbf{p}_{t-1}^i\|_2 > \lambda \cdot \bar{v}_t + \epsilon \tag{2}$$

is met, where λ is a scaling factor and ϵ is a minimum velocity margin to account for stationary or slow-moving states. This constraint ensures that individual track updates remain consistent with the global dynamics of the group.

3) State Update and Smoothing: Upon a successful association between track i and detection j^*, the position is updated using the measurement, while the velocity is updated using an exponential moving average to mitigate detection noise. The instantaneous velocity $\tilde{\mathbf{v}}_t^i$ is derived from the displacement $\tilde{\mathbf{v}}_t^i = \mathbf{d}_t^{j^*} - \mathbf{p}_{t-1}^i$. The smoothed velocity state is updated as

$$\mathbf{v}_t^i = \alpha \cdot \tilde{\mathbf{v}}_t^i + (1 - \alpha) \cdot \mathbf{v}_{t-1}^i, \tag{3}$$

where $\alpha \in [0, 1]$ is the smoothing factor ($\alpha = 0.1$). The position is updated to the measurement to prevent drift $\mathbf{p}_t^i = \mathbf{d}_t^{j^*}$.

4) Occlusion Handling (Coasting): In cases of missed detections or occlusions where track i is not associated with any detection, the system enters a "coasting" phase. The state is updated based purely on the motion model. This mechanism ensures the continuity of the trajectory through transient sensory gaps, maintaining the temporal consistency of the track until a new observation is associated. Tracks are terminated if they remain unassociated for $K_{\text{max}} = 3$ consecutive frames or if they exit the FOV.

5) Evaluation Metrics: To evaluate performance, the MOE tracker is compared to several state-of-the-art tracking algorithms CSRT [20], KCF [21], and NanoTracker [22] using the Sucess Rate (SR) metric. To create a reliable baseline for comparison, reference trajectories of the birds were manually annotated. For a sequence of N frames, SR is calculated for a single track as

$$\mathrm{SR} = \frac{1}{N}\sum_{t=1}^{N} \mathbb{1}(E_t < \gamma) \times 100\%, \tag{4}$$

where $\mathbb{1}(\cdot)$ is the indicator function, E_t is the location error of the track w.r.t. the manual annotation at timestep t, and γ is a threshold that we chose as $\gamma = 50\,\mathrm{px}$.

Initial evaluations were conducted using Seq. 3, Clip 6, focusing on the trajectories of 2 specific birds within a larger flock of 20. In the right camera where the birds initiate flight in close proximity, NanoTracker erroneously merges the trajectories, and the other trackers fail (see Fig. 5-left). The MOE system successfully navigates these complex maneuvers and occlusions, showing its reliability for this tracking task.

To further validate the robustness of these methods, they were stress-tested across 10 different simulated benchmarking trajectories that mimic extreme conditions, complementing the evaluations conducted on real-world data (see Fig 5-right, and the attached video). These results validate the MOE system as the superior tracking solution, demonstrating its consistent ability to maintain accurate tracking under demanding dynamics in both real-world and simulated environments, where baseline algorithms struggle and fail.

C. Stereo Matching and 3D Trajectories

To reconstruct full 3D trajectories of individuals in the flock from asynchronous event streams, a custom stereo-matching pipeline is implemented. Initial attempts utilized DDES [23], one of the first established methods for event-based stereo. While functional in standard scenarios, DDES proved insufficient for this experiment due to the high density of the flock and the rapid, overlapping movements of multiple agents, which created significant association ambiguities. Consequently, we developed a specialized approach that transitions from 2D pixel-space tracks in the left and right image to metric 3D coordinates using a multi-feature association cost and linear triangulation.

1) Image Rectification and Feature Extraction: Given the intrinsic and the extrinsic parameters obtained using

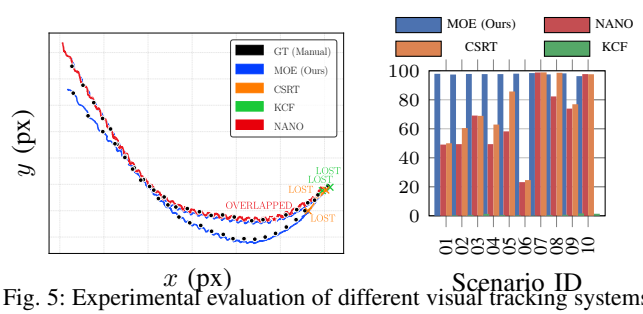

Fig. 5: Experimental evaluation of different visual tracking systems. *Left:* A qualitative comparison on Seq. 3, clip 6. *Right:* Success Rate (%) across 10 simulated scenarios.

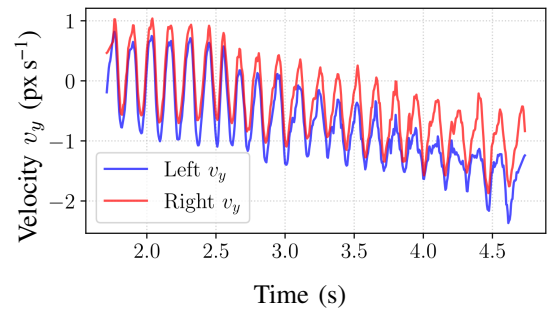

Fig. 6: Velocity signature correlation over time for a single bird.

the calibration procedure described in section III-C.2, the rectification transform and projection matrix are computed for each camera. Each track centroid (x, y) is transformed into the rectified space (x', y') such that the epipolar lines are horizontally aligned in both images, satisfying the constraint $y'_L \approx y'_R$, i.e. the y-coordinate of an object is almost equivalent in the left and right images. A Savitzky-Golay filter [24] with window length $n = 21$ is applied to the rectified coordinates to produce smoothed trajectories and instantaneous velocity vectors (v_x, v_y). This suppresses detection noise and provides a smooth signal for correlation analysis.

2) Multi-Feature Association: The correspondence problem is formulated as a global optimization task. For every pair of tracked trajectories from the left and right cameras that share a minimal temporal overlap, a cost function C_{total} is defined based on three physical constraints:

$$C_{\text{total}} = w_y \cdot \bar{E}y + w_m \cdot (1 - \rho_v) + w_s \cdot (1 - \rho_y), \tag{5}$$

where $\bar{E}_y$ is the mean vertical epipolar error, ρ_v and ρ_y represent correlation coefficients of the vertical velocities and trajectory shapes, and w_y, w_m, w_s are weight parameters.

The high temporal resolution of the event cameras is critical here, as it captures oscillations of the vertical velocity caused by wing-beat cycles with extreme precision. This temporal fidelity ensures that the motion profiles of the same individual are highly synchronized in the left and right images ($C_{\text{total}} = 0.9843$), as shown in Fig. 6. Solving the assignment problem using the Hungarian algorithm [25] finds the optimal set of matches by leveraging these unique biological motion signatures to resolve ambiguities even in dense flocks.

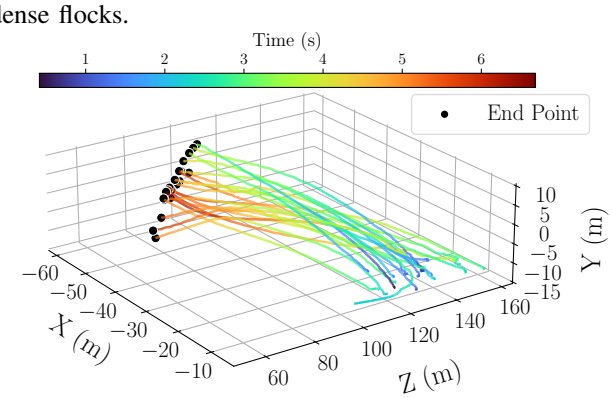

Fig. 7: Example reconstructed 3D trajectories of birds within the real-world flock.

To select the weights w_y, w_m, w_s, a ground truth dataset was established by manually matching trajectories across the stereo views based on visual inspection. Using two specific recording clips (Seq. 3, clip 6, and 11), the weight parameters were then manually tuned to maximize the assignment accuracy against this baseline. This empirical optimization yielded the final values of $w_y = 2$, $w_m = 40$, and $w_s = 20$.

3) Triangulation: For each matched pair of rectified centroids $\mathbf{x}'_L$ and $\mathbf{x}'_R$ at timestep t, the 3D position is calculated via linear triangulation. We solve the overdetermined system of equations $\mathbf{A}\mathbf{x} = \mathbf{0}$ for $\mathbf{x}$, where the matrix $\mathbf{A}$ is constructed from the left and right projection matrices $\mathbf{P}_L$, $\mathbf{P}_R$ as

$$\mathbf{A} = \begin{bmatrix} x'_L \mathbf{p}_L^{3\intercal} - \mathbf{p}_L^{1\intercal} \\ y'_L \mathbf{p}_L^{3\intercal} - \mathbf{p}_L^{2\intercal} \\ x'_R \mathbf{p}_R^{3\intercal} - \mathbf{p}_R^{1\intercal} \\ y'_R \mathbf{p}_R^{3\intercal} - \mathbf{p}_R^{2\intercal} \end{bmatrix}, \tag{6}$$

where $\mathbf{p}_i^n$ denotes the n-th column of the projection matrix i. The resulting 4D homogeneous vector $\mathbf{x}$ is normalized to yield the metric coordinates (X, Y, Z) in the reference camera frame. This pipeline enables the resolution of complex, intertwining flight paths within high-density flocks, as shown in the 3D reconstruction in Fig. 7.

VI. Dynamic Leadership

To validate the hypothesis that navigational authority in the flock is dynamic and spreads sequentially from bird to bird rather than resulting from a simultaneous, flock consensus, the temporal structure of collective turning maneuvers is analyzed.

A. Dimensionality Reduction and Heading Extraction

Observation of the recorded trajectories reveals that flight maneuvers are mostly spatially confined to a local plane specific to the concrete maneuver. This planar assumption is directly supported by previous studies on large-scale starling flocks, which also demonstrated that collective turns predominantly occur within a localized 2D plane [15]. Consequently, Singular Value Decomposition (SVD) was applied to the obtained 3D trajectories to project the motion onto a plane defined by the first two singular vectors $\mathbf{u}$, and $\mathbf{v}$. Within this local plane, the instantaneous heading for the i-th agent is computed as $\psi_i(t) = \arctan 2(\dot{v}_i, \dot{u}_i)$, where $\dot{u}_i$ and $\dot{v}_i$ are the agent's velocity components in the plane.

B. Temporal Hierarchy Detection

To identify the Turn Initiator (TI) and Turn Terminator (TT) individuals, we define a distinct behavioral event: the onset of a turn. A threshold $\theta_{\text{turn}} = 10°$ is established to distinguish intentional directional changes from sensor noise or minor stabilization adjustments. During a given maneuver, the turn initiation time t^i_{init} for bird i is identified as the earliest frame in which its heading deviates from the group's mean heading $\bar{\psi}_{\text{ref}}$ by more than the defined threshold θ_{turn}:

$$t^i_{\text{init}} = \min\{t \mid |\psi_i(t) - \bar{\psi}_{\text{ref}}| > \theta_{\text{turn}}\}. \tag{7}$$

The turn termination time is identified in a similar fashion but in reverse time. By sorting the set of initiation times $T = \{t^1_{\text{init}}, \ldots, t^N_{\text{init}}\}$ (and similarly for the termination times), the *temporal hierarchy* of the flock for that specific instance is reconstructed.

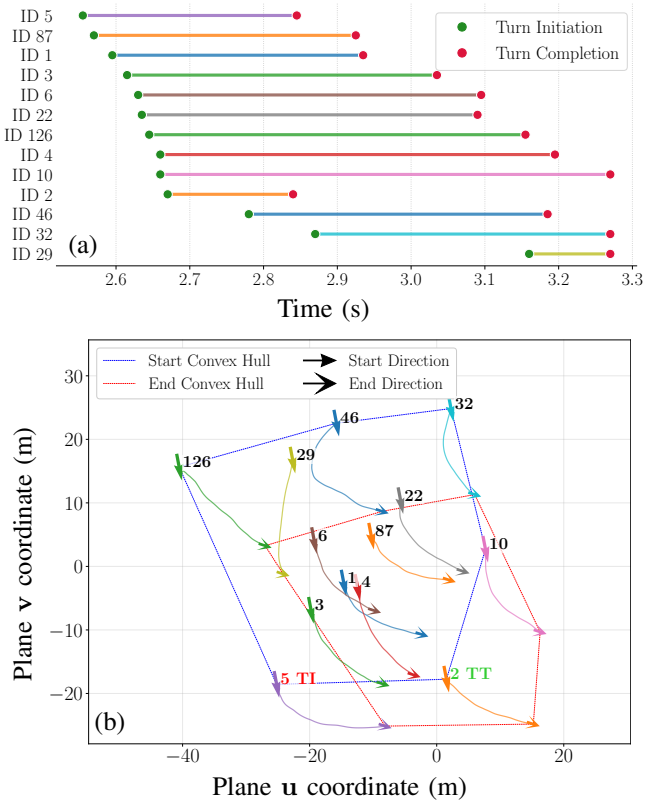

Fig. 8: Qualitative analysis of the collective decision-making process during a turn from Seq. 3, clip 006. (a) Temporal cascade of turn initiation and completion times. Pigeons IDs are extracted from the stereo pipeline and ordered chronologically to illustrate that individuals react sequentially to the maneuver. (b) Visualization of flight paths projected onto the principal motion plane, illustrating the spatial propagation of navigational authority.

C. Results: Asynchronous Propagation

Results of the data analysis are summarized in Fig. 8 and in Table III. The turn initiation sequences reveal a clear,

TABLE III: Evaluation of dynamic leadership during collective turns in real-world bird flocks.

Clip ID	**Turn** (θ)	**ID TI**	**ID TT**	Δt_{react} (ms)	**Plane Error** (m)	f_{wing} (Hz)	**Distance** (m)	**Num. IDs**	t_{total} (s)
Seq3-6	113.70°	**005**	**002**	164	1.188	4.32	103.179	20	7.12
Seq3-11	−136.37°	**001**	**007**	138	1.486	4.59	166.146	2	5.61
Seq3-15	−44.28°	**073**	**316**	145	0.681	4.70	67.117	88	4.61
Seq3-21	−78.82°	**001**	**035**	496	1.781	4.22	66.160	53	6.38

TABLE IV: Comparison of different control modes in simulations.

Clip	Mode	Dist. to path (m)	Traveled (m)	Dispersion (m)
006	1	31.92	210.45	19.18
	2	32.04	219.74	19.86
	3	28.54	184.70	19.89
011	1	47.00	202.60	5.77
	2	42.10	207.01	5.97
	3	27.31	201.77	6.29
015	1	49.45	268.50	39.07
	2	58.54	279.28	40.00
	3	44.82	222.47	37.49
021	1	41.13	358.62	33.64
	2	46.72	358.99	33.64
	3	44.40	310.23	33.81

Note: 1 = Leaderless Consensus, 2 = Fixed Leader, 3 = Dynamic Leader.

staggered cascade of reaction times rather than a clustered or simultaneous response (see Fig. 8 (a)). The data indicates that the bird identified as the topological leader (TI) initiates the maneuver, while the remaining flock members follow with a propagation delay. This strictly sequential reaction confirms that the flock operates on a perception-driven basis as the individuals do not respond to a global metric average. Instead, they react to localized visual cues of their immediate predecessors, such as optical expansion or orientation shifts.

The high temporal resolution of the event cameras allows for the quantification of such micro-delays in the bird motion, providing empirical evidence that directional decisions are seized by specific individuals and propagated through the flock. A critical finding enabled by the high temporal resolution is the dissociation between maneuver *initiation* and *termination*. While specific individuals trigger directional changes, they do not necessarily maintain navigational authority for the duration of the maneuver.

The maneuver's magnitude is captured by the total turn angle (θ), with the flock's flatness measured via the median distance to the fitted 2D plane, and the overall movement scale by the median 3D traveled distance. The bird-to-bird reaction delay (Δt_{react}) represents the average propagation time following the TI. We analyzed clips with varying group sizes (up to 88 individuals), confirming that this dynamic leadership strategy and its reaction patterns are scale-invariant. Additionally, the event cameras' high temporal resolution uniquely captured subtle biomechanical shifts, such as the flapping frequencies (f_{wing}) calculated from the fluctuations observed in the y-axis velocities, as shown in Fig. 6.

D. UAV Simulation

To evaluate the practical application of the observed biological principles in aerial swarm robotics, a comparative simulated experiment was designed using fixed-wing UAVs. Fixed-wing platforms were specifically chosen due to dynamical resemblance and their common non-holonomic constraints. The simulation framework evaluates the efficiency and spatial accuracy of the swarm across three distinct navigational paradigms: a representative consensus-based "boids" model, a static leader-follower model, and the proposed dynamic leadership model derived from empirical data. The simulation framework restricts agents to the biological counterpart's $316.4°$ FOV [26].

In the Leaderless Consensus model, each agent (denoted by i) follows desired velocity rules governed by

$$\mathbf{v}_{i,\text{des}} = w_{\text{coh}}\mathbf{v}_{\text{coh}} + w_{\text{rep}}\mathbf{v}_{\text{rep}} + w_{\text{align}}\mathbf{v}_{\text{align}} + w_{\text{tgt}}\mathbf{v}_{\text{tgt}} = \begin{bmatrix} 1/|\mathcal{N}|\sum_{j\in\mathcal{N}}(\mathbf{p}_j-\mathbf{p}_i)^T \\ 1/|\mathcal{N}|\sum_{j\in\mathcal{N}}(r_{\text{rep}}-||\mathbf{p}_j-\mathbf{p}_i||_2)\frac{(\mathbf{p}_j-\mathbf{p}_i)^T}{||\mathbf{p}_j-\mathbf{p}_i||_2} \\ 1/(|\mathcal{N}|+1)\left(\mathbf{v}_i+\sum_{j\in\mathcal{N}}\mathbf{v}_j\right)^T \\ \frac{(\mathbf{p}_{\text{tgt}}-\mathbf{p}_i)^T}{||\mathbf{p}_{\text{tgt}}-\mathbf{p}_i||_2} \end{bmatrix}^T \begin{bmatrix} w_{\text{coh}} \\ w_{\text{rep}} \\ w_{\text{align}} \\ w_{\text{tgt}} \end{bmatrix}, \tag{8}$$

where $\mathbf{v}_{\text{coh}}$ enforces flock cohesion, which introduces the concept of neighborhood ($\mathcal{N}$), that is spatially restricted by the agent's FOV. Furthermore, $\mathbf{v}_{\text{align}}$ serves to synchronize the agents' flight direction, $\mathbf{v}_{\text{rep}}$ ensures collision-free navigation by repulsing any individuals that enter a defined repulsion radius (r_{rep}), and, finally, navigates towards global spatial target using the directional vector $\mathbf{v}_{tgt}$. Weights associated to these terms are denoted accordingly as $w_{(\cdot)}$.

In the Fixed Leader mode, a single leader navigates toward the target while tethered to the swarm using the law:

$$\mathbf{v}^{\text{ldr}}_{i,\text{des}} = w_{\text{tgt}}\mathbf{v}_{\text{tgt}} + w_{\text{tether}}\begin{cases}\mathbf{v}_{\text{coh}} & \text{if } ||\mathbf{v}_{\text{coh}}||_2 \geq r_{\text{tether}} \\ 0 & \text{else}\end{cases}. \tag{9}$$

If the neighbors' center of mass exceeds a defined leash radius of $r_{\text{tether}} = 8\,\text{m}$, the tether mechanism activates to slow the leader down. Concurrently, the followers track the leader's position while avoiding collisions with other agents:

$$\mathbf{v}^{\text{follow}}_{i,\text{des}} = w_{\text{rep}}\mathbf{v}_{\text{rep}} + w_{\text{ldr}}\frac{\mathbf{p}_{\text{ldr}}-\mathbf{p}_i}{||\mathbf{p}_{\text{ldr}}-\mathbf{p}_i||_2}. \tag{10}$$

The Dynamic Leader mode uses eqs. (9), (10) but continuously elects the agent closest to the reference position as the new leader, utilizing a hysteresis threshold of $2\,\text{m}$ to prevent erratic switching.

In order to evaluate the performance of these methods without introducing parametrization-related bias, the controllers were optimized using the CMA-ES optimizer [27]. The optimization process tuned the maximum speed, the repulsion radius, and the weights of the velocity components $w_{(\cdot)}$. The algorithm minimized a comprehensive fitness function J enforcing realistic and safe behavior,

$$J = \lambda_1 e_{\text{com}} + \lambda_2\bar{d} + \lambda_3\bar{\dot{j}} + p_{\text{coll}}, \tag{11}$$

where e_{com} represents the macroscopic spatial tracking error between the swarm's center of mass and the trajectory executed by the center of mass of the biological flock, $\bar{d}$ is the average distance traveled by individuals, $\bar{\dot{j}}$ penalizes sharp velocity changes to minimize jitter, and p_{coll} applies a severe cost for breaching the minimum safety distance. The weighting coefficients $\lambda_1 = 10.0$, $\lambda_2 = 0.5$, and $\lambda_3 = 5.0$ are empirically set to prioritize macroscopic tracking while gently regularizing flight efficiency and heavily penalizing jitter. Meanwhile, the collision term $p_{\text{coll}} = 1000.0$ acts as a severe absolute penalty.

Initial positions and target waypoints are extracted directly from the empirical data (Seq. 3, clips 6, 11, 15, and 21), including number of agents.

The quantitative results in Table IV demonstrate the superiority of the dynamic leadership model. The classical consensus model struggled to adapt to rapid directional changes, resulting in trajectory deviations and increased travel distances. The static leader-follower approach often yielded the highest tracking errors and longest paths, as a single fixed leader cannot maintain an optimal position during sharp maneuvers. Conversely, the dynamic leadership model consistently recorded the shortest travel distances and the lowest deviations from the true flight path. This proves the dynamic paradigm is both the most energetically efficient and the most accurate representation of biological pigeon flock dynamics.

VII. Conclusion

This work advances the foundations of swarm robotics by investigating the interaction mechanisms underlying the collective behavior of biological flocks. We introduced a stereo event-camera system capable of capturing data at an exceptionally high temporal resolution, together with an automated pipeline for reconstructing individual 3D trajectories within large-scale, densely packed flocks. The resulting dataset of rock doves (*Columba livia*) provides actuation-level detail, enabling quantitative behavioral analysis at an unprecedented temporal fidelity. Analysis of the spatio-temporal characteristics of multiple flock maneuvers supports the hypothesis of dynamically evolving navigation authority within the group. Building on this insight, we implemented a dynamic leadership strategy in a multi-robot simulation framework. The proposed approach demonstrated improved efficiency in coordinated turning maneuvers compared to static leader-based and consensus-based control strategies. Moreover, the simulated fixed-wing UAV trajectories exhibited the highest similarity to the biological flight patterns when governed by dynamic leadership, further supporting the presence of adaptive hierarchical organization in bird flocks. The observed performance gains in multi-UAV systems highlight the practical value of biologically grounded interaction models. The presented dataset and methodology open new opportunities for high-fidelity investigation of collective behavior and its principled transfer to robotic swarm systems.

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